

CONTACT INFORMATION	Department of Statistical Science Baylor University One Bear Place 97140 Waco, Texas 76798, USA	Phone (480) 320-8898 E-mail carsonslater7@gmail.com WWW https://carsonslater.github.io
RESEARCH INTERESTS	Time series clustering, anomaly detection, and nonhomogenous hidden markov models with applications in urban water distribution and advanced metering infrastructure technologies.	
EDUCATION	Baylor University PhD Statistical Science Advised by Amanda S. Hering. Research involves developing statistical methods for time series clustering with applications in urban water distribution.	2024–2027 [Expected]
	Baylor University M.S. Statistical Science	2023–2024
	Wheaton College (IL) B.S. Applied Mathematics with Statistics B.A. Economics Edman Presidential Scholarship	2019–2023
PROFESSIONAL EXPERIENCE	Baylor University Statistical Consultant Consulting for ongoing interdisciplinary research in the computer engineering and environmental science departments, and for the American Holistic Nursing Association. Currently developing analytical tools for the Baylor Athletics department with recent developments in NIL.	August 2024–Present
	Research Assistant Developing statistical methods for urban water distribution processes, leveraging unsupervised learning on time series for customer-level water utility data. Research is funded by Denver Water.	January 2024–Present
	Wheaton College Mathematics Teaching Assistant Facilitated help sessions for undergraduate students in the topics of Machine Learning and Probability Theory.	August 2022–May 2023
	Wheaton Academy Center for Lifelong Learning Academic Coach Mentored students 1-on-1 to sharpen math skills from Algebra I to AP Statistics.	January 2023–March 2023
ARTICLES IN PREPARATION	Carson Slater , Ryan Shepler, Samantha Bryan, Tzahi Y. Cath, and Amanda S. Hering. 2026. “Identifying Outdoor Watering Events with High-Frequency Metering Data to Assess Residential Water Consumption”. <i>In Preparation</i> .	
PEER-REVIEWED PUBLICATIONS	Brittany J. Carnathan, Dinny Stevens, Swarna Shikha, Carson Slater , Nathen Byford, Rodney X. Sturdivant, Kuzy Zarzosa, W. Evan Braswell, and Christie M. Sayes. 2025. “Assessing the Effects of Surface-Stabilized Zero-Valent Iron Nanoparticles on Diverse Bacteria Species	

Using Complementary Statistical Models”. *Journal of Functional Biomaterials* 16 (3). <https://doi.org/10.3390/jfb16030113>

CONFERENCE PROCEEDINGS & OTHER ARTICLES

Carson Slater, Ryan Shepler, Samantha Bryan, Tzahi Y. Cath, and Amanda S. Hering. 2026. “Identifying Outdoor Watering Events with High-Frequency Metering Data to Assess Residential Water Consumption”. 2026 . <https://doi.org/10.1109/WMCS62019.2024.10619025>

Jonathan E. Swindell, **Carson Slater**, Samuel Hussey, Charles Baylis, and Robert J. Marks. 2024. “Assessing Interference with Regression Analysis Techniques”. *2024 IEEE Texas Symposium on Wireless and Microwave Circuits and Systems (WMCS)*. <https://doi.org/10.1109/WMCS62019.2024.10619025>

Amanda S. Hering, Douglas Nychka, Luke Durell, **Carson Slater**, Michael Poor, Greg Hamerly, and Tzahi Y. Cath. 2024. “Mo(Wa)²TER Introduction to Data Science Course Material”. *Harvard Dataverse*. <https://doi.org/10.7910/DVN/BZJW2E>

Carson Slater and Ying Li. 2022. “Forecasting Fast-Moving Consumer Goods with Generalized Additive Models”. *2022 University of Illinois Chicago Undergraduate Mathematics Symposium*. (Poster Presentation)

HONORS & AWARDS

Wheaton College Athletics, NCAA

2019-22

Varsity Baseball Player

Practiced baseball six days a week for multiple hours and played over forty-games during the spring semester.

Duke/North Carolina State University Summer Institute for Biostatistics

2022

Participant

Engaged standard biostatistical methods; Hackathon studying myocardial infarction.

Valley Christian High School, Chandler AZ

2019

Christian Character Award

Given to one graduating senior each year that exemplified honorable character in their tenure at the high school.

SERVICE & VOLUNTEER

Křesťanská Akademie Mladých (CZ)

June 2023–August 2023

Volunteer Intern

Assisted local churches by teaching English and sports in schools and camps throughout the Czech Republic.

SKILLS

Technology

Docker, Positron ($\approx$ VS Code), Git/Github, reveal.js, R/RStudio, SQL, duckDB, L^AT_EX.

Python

pandas, numpy, PyTorch (Lightning), plotnine, seaborn, plotly, scikit-learn, Jupyter

R

tidyverse (dplyr, ggplot2, etc.), TensorFlow (Keras), torch, shiny, RMarkdown/Quarto, rstan, rjags, caret, tidymodels, targets (gittargets, proffer, etc.)

Statistics

statistical modeling, hypothesis testing, mixture models, time series, regression, linear mixed models, clustering, classification, deep learning, PCA, random forests, ensemble models (e.g., XGBoost), SVM, Bayesian methods, spatiotemporal modeling